

TutorPlus AI - Complete Testing Checklist

Pre-Testing Setup

- ☐ Backend running: `python run.py` (`http://localhost:8000`)
 - ☐ Frontend running: `npm run dev` (`http://localhost:5173`)
 - ☐ Database initialized (check `/data/tutorplus.db` exists)
 - ☐ ChromaDB initialized (check `/data/chroma_db` exists)
 - ☐ All dependencies installed
 - ☐ Environment variables configured (.env files present)
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1. AUTHENTICATION TESTING

1.1 Registration

- ☐ Register with valid email, username, password
 - Expected: User created, redirect to login
 - Check: User appears in database
- ☐ Register with existing email
 - Expected: Error message "Email already exists"
- ☐ Register with existing username
 - Expected: Error message "Username already taken"
- ☐ Register with short password (< 8 chars)
 - Expected: Validation error
- ☐ Register with invalid email
 - Expected: Validation error
- ☐ Register with all languages (en, yo, ha, ig)
 - Expected: Each registers successfully

1.2 Login

- ☐ Login with correct email and password
 - Expected: Token generated, redirect to dashboard
 - Check: Access token stored in localStorage
- ☐ Login with incorrect password
 - Expected: Error message "Invalid email or password"

- ☐ Login with non-existent email
 - Expected: Error message "Invalid email or password"
- ☐ Login with empty fields
 - Expected: Validation error
- ☐ Token persistence (refresh page after login)
 - Expected: Still authenticated, dashboard shows user info

1.3 Authentication Flow

- ☐ Access dashboard without login
 - Expected: Redirect to login page
 - ☐ Access protected routes without token
 - Expected: Redirect to login
 - ☐ Token refresh (wait for token to expire)
 - Expected: Automatic refresh, stay logged in
 - ☐ Logout functionality
 - Expected: Tokens cleared, redirect to login
 - ☐ Access protected route after logout
 - Expected: Redirect to login
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2. DASHBOARD TESTING

2.1 Dashboard Display

- ☐ Dashboard loads successfully when authenticated
- ☐ Welcome message shows user's name
- ☐ All feature cards display (Tutor, MCQ, Progress, Voice)
- ☐ Feature cards are clickable and navigate correctly
- ☐ Quick stats section shows (0 for new user)
- ☐ Header shows logged-in user info
- ☐ Logout button works and clears session

2.2 Navigation

- ☐ "Tutor Chat" card → TutorPage
- ☐ "MCQ Practice" card → MCQPage
- ☐ "Progress Tracker" card → ProgressPage

- ☐ "Voice Chat" card → TutorPage with voice mode
 - ☐ Back navigation works from all pages
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3. TUTOR INTERFACE TESTING

3.1 Chat Functionality

- ☐ Load TutorPage successfully
- ☐ Input field accepts text
- ☐ Send button is disabled when input is empty
- ☐ Message sends and appears in chat
- ☐ AI response loads and displays
- ☐ Chat history persists (multiple messages)
- ☐ Auto-scroll to latest message

3.2 Settings

- ☐ Subject field accepts input
- ☐ Language dropdown has all 4 options
- ☐ Changing language affects response language
- ☐ Settings persist in session

3.3 Voice Features (When Google Cloud is configured)

- ☐ Voice input button appears
- ☐ Audio playback works for AI responses
- ☐ Play button shows in message with audio
- ☐ Multiple messages can have audio

3.4 Error Handling

- ☐ Send message with no subject
 - Expected: Response still generated
 - ☐ Network error during message send
 - Expected: Error message displayed, retry possible
 - ☐ Empty message send attempt
 - Expected: Send button disabled
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4. MCQ TESTING

4.1 MCQ Generation

- ☐ Navigate to MCQ Page
- ☐ Generate MCQ without subject
 - Expected: Alert "Please fill in subject and topic"
- ☐ Generate MCQ with subject and topic
 - Expected: Questions display, loading shows during generation
- ☐ Change difficulty (easy, medium, hard)
 - Expected: Generates questions of selected difficulty
- ☐ Change number of questions (1-20)
 - Expected: Correct number of questions generated
- ☐ Change language (en, yo, ha, ig)
 - Expected: Questions in selected language

4.2 MCQ Solving

- ☐ Questions display with options A, B, C, D
- ☐ Can select different answers for each question
- ☐ Selected answer is highlighted
- ☐ Can change answer before submission
- ☐ Submit button works after selecting answers
- ☐ Submit disabled until at least one answer selected
- ☐ Success message appears after submission

4.3 MCQ Mode Switching

- ☐ Toggle between "Generate MCQ" and "Solve Questions"
 - ☐ Switch modes without losing data
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5. PROGRESS TRACKING TESTING

5.1 Progress Page Load

- ☐ ProgressPage loads successfully
- ☐ Loading indicator appears initially
- ☐ Stats cards display (Total Lessons, Subjects, Avg Score)

- ☐ Subject buttons display all available subjects

5.2 Progress Filtering

- ☐ Click subject button to filter progress
- ☐ Selected subject is highlighted
- ☐ Progress items filtered by subject
- ☐ Click again to deselect and show all
- ☐ Progress bars display correctly

5.3 Progress Display

- ☐ Progress items show subject name
- ☐ Progress items show topic name
- ☐ Quiz scores display with correct color coding
 - 80+: Green
 - 60-79: Yellow
 - <60: Red
- ☐ Progress bars fill correctly based on completion
- ☐ Lesson count shows "X of Y lessons completed"

5.4 Empty State

- ☐ When no progress exists, empty state message shows
 - ☐ Empty state is user-friendly
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6. RAG & CURRICULUM TESTING (Admin)

6.1 Curriculum Upload (if Admin feature is added)

- ☐ Navigate to admin panel
- ☐ Upload PDF file
 - Expected: File selected, upload starts
- ☐ Invalid file type (e.g., .txt)
 - Expected: Error message
- ☐ Large PDF file (>10MB)
 - Expected: Upload completes without timeout
- ☐ Check curriculum stats endpoint

- Expected: Chunk count shows correctly

6.2 RAG Integration with Tutor

- ☐ Ask question without curriculum loaded
 - Expected: Generic response or "No curriculum found"
 - ☐ Upload curriculum for subject
 - ☐ Ask question about uploaded subject
 - Expected: Response uses curriculum data
 - ☐ Verify sources appear in response
 - Expected: Sources metadata displayed
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7. API INTEGRATION TESTING

7.1 Authentication Endpoints

- ☐ POST /api/auth/register → Returns user data
- ☐ POST /api/auth/login → Returns tokens
- ☐ GET /api/auth/me → Returns current user
- ☐ POST /api/auth/refresh → Returns new access token
- ☐ POST /api/auth/logout → Success response

7.2 Tutor Endpoints

- ☐ POST /api/tutor/ask → Returns response with text
- ☐ POST /api/tutor/generate-mcq → Returns MCQ questions
- ☐ POST /api/tutor/generate-voice → Returns audio URL
- ☐ POST /api/tutor/process-voice-query → Transcribes audio (when enabled)

7.3 Progress Endpoints

- ☐ GET /api/progress/my → Returns user progress list
- ☐ POST /api/progress/track → Creates/updates progress
- ☐ GET /api/progress/student/{subject} → Returns subject progress
- ☐ PUT /api/progress/student/{subject} → Updates subject progress

7.4 MCQ Endpoints

- ☐ GET /api/mcq/{id} → Returns specific MCQ
- ☐ GET /api/mcq/subject/{subject} → Returns MCQs by subject

- ☐ POST /api/mcq/attempt → Submits answer attempt
- ☐ GET /api/mcq/attempts/my → Returns user attempts

7.5 Admin Endpoints

- ☐ POST /api/admin/upload-curriculum → Uploads PDF
 - ☐ GET /api/admin/curriculum-stats → Returns chunk count
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8. UI/UX TESTING

8.1 Responsive Design

- ☐ Desktop (1920x1080) - All layouts correct
- ☐ Tablet (768x1024) - Grid adjusts correctly
- ☐ Mobile (375x667) - Single column layout
- ☐ Mobile landscape (667x375) - Readable

8.2 Visual Elements

- ☐ All Lucide icons load and display
- ☐ Tailwind colors apply correctly
- ☐ Cards have proper shadows
- ☐ Buttons have hover effects
- ☐ Input fields have focus states
- ☐ Loading spinners animate

8.3 Accessibility

- ☐ Tab navigation works
 - ☐ Form labels associated with inputs
 - ☐ Color contrast meets WCAG standards
 - ☐ Alt text for images (if any)
 - ☐ Error messages clear and descriptive
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9. PERFORMANCE TESTING

9.1 Loading Times

- ☐ Dashboard loads < 2 seconds

- ☐ TutorPage loads < 1 second
- ☐ MCQPage loads < 1 second
- ☐ ProgressPage loads < 2 seconds
- ☐ AI response generates within 5-10 seconds

9.2 Memory Usage

- ☐ Chat history doesn't cause slowdown
- ☐ Multiple messages don't freeze UI
- ☐ Page refresh clears memory

9.3 Network

- ☐ Slow network (3G) still functional
 - ☐ Request timeout handling works
 - ☐ Retry mechanism works
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10. ERROR HANDLING TESTING

10.1 Network Errors

- ☐ Disconnect during API call
 - Expected: Error message, option to retry
- ☐ Network timeout
 - Expected: Timeout error, retry available
- ☐ Server error (500)
 - Expected: User-friendly error message

10.2 Validation Errors

- ☐ Empty form submission
 - Expected: Validation messages
- ☐ Invalid email format
 - Expected: Validation error
- ☐ Character limits
 - Expected: Field limits respected

10.3 State Errors

- ☐ Close browser tab during action
 - Expected: Session continues in other tab
 - ☐ Rapid clicking (multiple sends)
 - Expected: Only one request sent
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11. BROWSER COMPATIBILITY TESTING

- ☐ Chrome (latest)
 - ☐ Firefox (latest)
 - ☐ Safari (latest)
 - ☐ Edge (latest)
 - ☐ Mobile Safari (iOS)
 - ☐ Chrome Mobile (Android)
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12. DATA PERSISTENCE TESTING

12.1 Database

- ☐ New user data saved correctly
- ☐ User can log back in after close
- ☐ Progress data persists
- ☐ MCQ attempts saved
- ☐ No data loss on refresh

12.2 Local Storage

- ☐ Tokens stored correctly
 - ☐ Tokens cleared on logout
 - ☐ localStorage doesn't exceed limit
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13. MULTILINGUAL TESTING

- ☐ Register with each language preference
- ☐ AI responds in selected language (en, yo, ha, ig)

- ☐ UI language preference affects backend responses
 - ☐ Language switching works seamlessly
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14. SECURITY TESTING

- ☐ No hardcoded credentials in frontend
 - ☐ API keys not exposed in browser
 - ☐ Tokens in secure HttpOnly cookies (if applicable)
 - ☐ No sensitive data in localStorage
 - ☐ CORS properly configured
 - ☐ Password not echoed in network requests
 - ☐ SQL injection protection (backend)
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15. END-TO-END FLOWS

15.1 Complete User Journey

- ☐ New user: Register → Login → Dashboard
- ☐ Ask Question: Dashboard → Tutor → Ask → Get Response
- ☐ Practice MCQ: Dashboard → MCQ → Generate → Solve → Submit
- ☐ Track Progress: Dashboard → Progress → View Stats
- ☐ Logout: Any page → Logout → Login page

15.2 Real-World Scenarios

- ☐ User takes 30+ message chat
 - ☐ Generate 20 MCQ questions
 - ☐ Solve 50+ MCQ questions
 - ☐ Switch between pages multiple times
 - ☐ Upload large curriculum (>5MB)
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16. TEST RESULTS SUMMARY

Component	Status	Notes
Authentication	[] Pass / [] Fail	

Component	Status	Notes
Dashboard	☐ Pass / ☐ Fail	
Tutor Interface	☐ Pass / ☐ Fail	
MCQ System	☐ Pass / ☐ Fail	
Progress Tracker	☐ Pass / ☐ Fail	
API Integration	☐ Pass / ☐ Fail	
UI/UX	☐ Pass / ☐ Fail	
Performance	☐ Pass / ☐ Fail	
Error Handling	☐ Pass / ☐ Fail	
Security	☐ Pass / ☐ Fail	
OVERALL	☐ Pass / ☐ Fail	

Known Issues & Notes

- Issue #1: _____
- Issue #2: _____
- Note #1: _____
- Note #2: _____

Sign-Off

- Tested by: _____
- Date: _____
- Environment: _____
- Browser: _____
- Status: ☐ Ready for Production / ☐ Needs More Testing

Post-Testing Checklist

- ☐ All critical issues resolved
- ☐ All non-critical issues documented
- ☐ Performance meets requirements
- ☐ Security reviewed
- ☐ Code reviewed
- ☐ Documentation updated
- ☐ Ready for deployment